

Chapter 1: Foundations of Machine Learning

1.1 Introduction

Machine Learning is defined as the scientific study of algorithms and statistical models that computer systems use to perform a task without being explicitly programmed to do so.

For example, spam filtering in email systems classifies messages based on historical patterns.

1.2 Linear Regression and Optimization

The objective function in ordinary least squares minimization is expressed as:

$$J(w) = \frac{1}{2m} \sum (h_w(x) - y)^2$$

Model Performance Comparison:

Model	Accuracy	Latency_ms
LinearRegression	0.84	1.2
RandomForest	0.92	8.5
NeuralNetwork	0.96	15.0